

Whisky

J.R. Piggott

FORMERLY UNIVERSITY OF STRATHCLYDE, GLASGOW, SCOTLAND

15.1 Introduction¹

Whiskies are distilled beverages which have been produced by the distillation of fermented cereals and have been matured in wooden casks. They can be classified on the basis of the nature of the cereal, the manner in which the product is blended, if at all, and the country of origin. Although there is some evidence for a distillation-type process in Egypt, the earliest reasonably reliable report is in Salerno in about 1100 AD [1]. The first reports of whisky production arise in Ireland in the twelfth century, and the first report of whisky production in Scotland is found in records dating from 1494 [2]. The term ‘whisky’ was derived from the Gaelic *uisge beatha* which means ‘water of life’. Major countries of production are Scotland, the USA, Ireland, Canada and Japan. The top 10 whisky markets by volume are India, USA, France, Japan, United Kingdom, Spain, Canada, Brazil, South Africa, and Germany.

Regulations in the European Union (EU) define whisky as a spirit produced by the distillation of a mash of cereals and matured for at least 3 years in wooden casks [3]. There are limitations on the enzymes which can be used to hydrolyze starch to fermentable sugars and on the distillation strength, and yeast must be used for the fermentation. Within this definition, Scotch and Irish whiskeys are further restricted [4]. In practice this provides for three distinct products: malt whisky, produced from 100% malted barley; grain whisky, produced from unmalted cereal grains; and the bulk of the whisky produced in Scotland which is blended from 60% to 70% grain whisky and 30–40% malt whiskies. This blended whisky usually contains up to 40 individual malts which are blended to produce a consistent brand flavor, and every component of the blend must be matured for the minimum period, or the age specified on the bottle. Malt whiskies may contain either a blend of malt whiskies of a given minimum age, or malt whisky from a single distillery. Scottish malts have a different character depending upon whether they are from the west coast, such as Campbeltown and the Islay region, which produce highly peated malts, or

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from the Highlands or the Lowlands. Irish whiskies have a similar flavor to Scotch but do not have a peaty character.

Whisky produced in the USA has evolved in a different legislative framework, allowing for a greater variety of distinct products [5]. The basic definition of US whiskey is that it is distilled from a fermented grain mash at less than 95% alcohol (190° US proof). Within this broad definition there are two special categories which are bottled as such or provide flavoring whiskies for blending. In the first group, the cereal content of the mash is restricted to not less than 51% of the named cereal (e.g., rye for rye whiskey, maize for Bourbon), the distillation must be at less than 80% alcohol, and maturation must be in new charred oak. Bourbon takes its name from Bourbon County in Kentucky. Corn whiskey must be from a mash of 80% maize, and matured in new uncharred or used oak. If matured for two or more years, these are 'straight whiskies'. They are to the blender the equivalent of the Scotch malts, and provide the majority of the flavor in a blend with lighter flavored grain spirits. US blenders are also permitted to use 'blending materials', such as sherry or blending wine, at up to 2.5%.

Canadian whiskey production follows the same general practice as in Scotland and the USA. Canadian blends are light column still whiskies with normally 10–15% of heavier-bodied flavoring whiskies. Up to 9.09% (on the basis of absolute alcohol content) of 'flavourings' is permitted, which may include wines and sherries, rum, brandy, Bourbon or malt whisky [6]. Maturation must be for not less than three years, and oak casks of 182 L (40 imperial gallons) are commonly used. The range in flavor required by the blender cannot be achieved by using the products of many different distilleries, but techniques such as variation of the composition of the mash (corn, wheat, rye or barley), cooking procedures, yeast strain, distillation procedure, barrel type and maturation period can be used. Traditionally, pot stills were used but these have largely been phased out.

Similar products are produced in many other countries [7], and are subject to local regulations which, while generally similar in principle to the EU definition, allow different extents of variation. In Brazil, regulations permit the addition of a proportion of agricultural alcohol to blended whisky, and in Japan locally produced spirit may be blended with imported Scottish malts. Spain is a substantial producer of whisky which must meet the EU requirements. The majority of whisky consumed in India is "Indian-made foreign liquor". These include whisky, rum and brandy, but Indian-made foreign liquors are generally manufactured from molasses, and therefore do not meet the standards commonly accepted for whiskies. EU and USA regulations also include provision for grain spirits or grain brandies which are essentially similar to whiskies but not matured.

15.2 Raw Materials

The major grains used in whisky production are maize (*Zea mays*), barley (*Hordeum polysticum*), wheat (*Triticum vulgare*) and rye (*Secale montanum*). These are the grains traditionally used for whisky, on the basis of a high fermentable starch content providing

a good yield of alcohol [8]. In Scotland and Ireland, all the amylase activity required for starch breakdown is obtained from malted barley, either in the form of a kilned malt which is stored at a moisture content of 5%, or a green malt which is prepared specifically on site as a source of enzyme activity for production of grain whisky. Some regulations also permit the use of other (normally microbial) enzymes during wort production. Cultivars are normally selected on the basis of extract and fermentability, corresponding to spirit yield, or enzyme activity in the case of malt for grain distilling. Enzyme activity should be considered under process conditions, as this will not be the same as activity measured under optimum laboratory conditions. Ethyl carbamate is an undesirable trace component in distilled beverages, and its content in malt whisky is largely determined by the cyanogenic precursor, epi-heterodendrin, in malt. Barley cultivars that do not contain the precursor are used. The gene responsible for its production has been identified, and differences in the DNA sequence between producers and non-producers (i.e., different alleles) have been used to develop a simple diagnostic assay that discriminates between them [9]. This can be used as a screening test to eliminate experimental varieties which are producers.

Water is required for mashing, cooling during distillation, dilution of the spirit before maturation, and blending. The quality of the product may be influenced by the nature of the water supplied, and many distilleries take their water supply from individual water courses or special wells. Soft water is normally used for malt distilleries, but some degree of hardness is preferred in grain distilleries since this has a stabilising effect on the amylase enzymes. The water should be monitored for chemical and microbiological contamination and be free of decaying vegetable matter. The water used for dilution of the spirit must have low calcium and iron content, and is normally demineralised, to minimise the risk of discoloration or precipitation occurring in the final product.

The yeast used for fermentation is normally specially grown distilling yeast which has good physiological and microbiological properties for producing alcohol [10]. However, some recycled brewer's yeast may be used in Scotland and yeasts from previous distillery fermentations may be used in some Bourbon processes; both influence the range of flavor products in the wash. The main criterion for yeast strain selection is normally efficiency in converting fermentable sugars to ethanol. To a limited extent strains may be selected to provide the required composition and flavor characteristics in the distillate, particularly the ester content, and yeast cultivation and storage conditions also affect its performance. Yeasts expressing glucoamylase activity have been developed.

Barrels used in whisky production are produced from either American white oak, *Quercus alba* and related species, or the Spanish oak, *Quercus robur* [11]. Scotch whisky is traditionally matured in barrels which have previously been used for Bourbon maturation or sherry production. Spanish sherry production is carried out in barrels constructed from either Spanish or American white oak. Except where the regulations provide otherwise, barrels are commonly reused until they fail to provide a satisfactory maturation. Such barrels may be regenerated by scraping out the inner surface and charring the inside of the barrel, but this does not produce performance equivalent to a

new barrel. During periods when sherry casks have been in short supply, barrels have also been wine treated or wine seasoned (filled with sherry for up to 6 months), and Madeira, port or ale casks are also occasionally used.

15.3 Production

15.3.1 Malting and Mashing

The traditional method of production involves the production of wort from cereal grains using amylases from barley [12]. These α - and β -amylases are produced by the barley during the malting process. Malting of barley is carried out by steeping the grain in water for 2–3 days and permitting the grain to germinate until the root is approximately one third the length of the grain. When this stage is reached the grain is dried down to a moisture level of 5% by kilning. The grain is then stored until immediately prior to use when it is milled before being transferred to a mash tun. In Scotch whisky production, peated malt is prepared by drying the malt in the presence of smoke from a peat fire and is an important source of flavor compounds, normally assessed on the basis of the total phenols content of the malt (up to 50 ppm), although other components may contribute to flavor. Nitrosamine contamination of malt could arise from the use of natural gas in directly fired kilns but is controlled by the use of burners designed to reduce nitrogen oxides, by burning sulfur to add sulfur dioxide to the air stream, or by indirect firing. Most of the breakdown of the starch occurs during the process of mashing which is carried out in a mash tun. Barley grist can be mashed directly since it has a high amylase activity, whereas worts produced from maize and other cereals require the addition of malted barley as a source of enzymes. The process of mashing may be a simple batch process similar to the infusion mashing techniques used in brewing beer, in which the grist is mixed with water at 63–68°C for 0.5–1.5 h before the wort is filtered off. However, in the production of grain whisky [13], an initial cooking stage, to gelatinise the starch of the non-barley cereals, is required. The cooking may be carried out as a batch process at 120°C for 1.5 h, or, as is more usual in North America, as a continuous process. In either case the mash is cooled to 60–65°C before the addition of the malted barley. Careful control is required at this stage to prevent the solidification of the gelatinised starch. When a cooking stage is used, amylase activity is normally provided by adding 10–15% freshly malted barley (green malt) after the cooking stage.

During the malting stage, β -glucanase enzymes are also produced; these facilitate the breakdown of the endosperm cell walls and the release of starch and proteases. The latter are involved in the breakdown of some of the barley grain proteins to amino acids and peptides. These amino acids are an essential nitrogen source for the growth of the yeast.

The mash is usually filtered prior to the fermentation to give a clear wort, but in some processes a higher yield is obtained by fermenting the whole mash directly.

15.3.2 Fermentation

The wort, which is produced when the contents of the mash tun has been filtered, is then transferred to a fermentation vessel and fermented using one or more strains of the yeast *Saccharomyces cerevisiae* [10,14]. When the wort is not boiled (as is the case in malt whisky production), it contains micro-organisms and active amylases. It is cooled to 20–25°C before inoculation with a strain of distilling yeast and, in some processes, secondary yeasts. Yeast growth occurs rapidly and is completed in the first 8–12 h. Ethanol production follows growth initially, then continues in a linear manner until the metabolisable sugars have been consumed. The resulting solution contains approximately 7% ethanol and a large number of flavor compounds. During the growth period, wort sugars are rapidly metabolised and the temperature may rise to 33°C if it is not controlled. The pH drops from an initial value of 5.0–5.5 to 4.2–4.5 as a result of the production of organic acids, such as acetate, succinate and pyruvate. A wide range of flavor compounds, such as higher alcohols, fatty acids, esters and carbonyl compounds, are also produced throughout the fermentation; they pass through into the distillate and constitute an important part of the product. In contrast, glycerol, which is also produced, is non-volatile and does not pass into the distillate; this represents an important loss of yield. At the end of the fermentation, usually between 48 and 72 h, death and autolysis of the yeast may occur. The fermentation is not a sterile process, so bacterial contamination can occur. Most of the bacteria which are introduced from the raw materials die during the early stages of the fermentation. However, lactobacilli can survive and grow with a concurrent formation of lactic acid during the later stages of the fermentation. Limited growth of lactic acid bacteria may be beneficial in flavor terms [15], but to prevent excessive growth and resulting yield losses fermentation is normally completed in 40–48 h, rather than the longer times traditionally used. Some distilleries may use longer fermentation times to accommodate work patterns. Heavy contamination by lactic acid bacteria can also result in a loss of ethanol yield and the production of acrid off-flavours.

15.3.3 Distillation

The fermented wash or beer (wort after fermentation) is then distilled either in a series of two or three pot stills, or in a series of column stills. The distillation process varies with the type of product and the country of origin. Historically, the first whisky processes used copper pot stills and this type of process is still used for the production of malt whisky in Scotland, where two stills - the wash still and the spirit still - are used. In Ireland, a three-still process is typical. Pot stills have also been used in the USA and Canada, but grain spirit is normally produced by continuous distillation in column stills. The first continuous process is associated with Scotch whisky production and the development of the Coffey or Patent still by Aeneas Coffey in 1830 [16]. The ethanol concentration of the distillate determines the extent of impurities (congeners) present, a higher strength distillate being purer aqueous ethanol with a less intense flavor.

In batch distillation processes [17], all the contents of the fermenter are transferred to the large wash still and boiled for 5–6 h until all the ethanol has been recovered, to produce a distillate, low wines, which contains 20–25% (v/v) ethanol. This is transferred to the low wines or spirit still where it is distilled to give a spirit containing around 70% (v/v) ethanol. In Scotch malt whisky production, the first fraction of the distillate, and the third fraction (after collection of the spirit) are recycled for redistillation with a subsequent batch of low wines. The Coffey still consists of two columns, referred to as the analyser or beer column, and the rectifier column [18]. Wash is fed toward the top of the first column and alcohol-free stillage is withdrawn at the bottom of the column; vapors are fed from the top to the base of the rectifying column. It produces a 190° US Proof spirit and is still used for the production of grain spirit in Scotland. In the USA, whiskey production employs more complex distillation processes involving the use of up to five columns. The heat for most distillation processes is supplied as steam at the present time, although pot stills directly fired by solid fuel or gas are also in use. In addition to producing spirit, distillation processes produce a large quantity of stillage which is processed into various by-products, such as dark grains and distiller's solubles, for animal feed.

15.3.4 Maturation

All whiskies are matured and most legal definitions of whisky incorporate a minimum maturation period. The legal minimum varies from 1 year for some American grain whiskies to 3 years for Scotch, Irish and Canadian whiskies. In effect these are the minimum requirements and most spirits are matured for longer periods. Branded products are generally matured for between 4 and 8 years, while single malts and premium straight whiskies are matured for eight to more than 18 years. The process of maturation involves storing the raw spirit from the still, diluted to 50–80% ethanol, in barrels made from a limited range of oak woods [11]. The type of cask used greatly affects the properties of the mature spirit. American straight whiskies (e.g., straight Bourbon whiskey, but excluding straight corn whiskey) are matured in new charred American oak barrels that by law cannot be reused. Maturation in new charred casks is associated with an increase in color, solids, acids, esters, aldehydes and tannins, together with increases in sugars, such as arabinose, glucose and xylose. New charred casks give a strong sweet, wood-derived flavor to the final product. The casks used for straight whisky maturation are American standard barrels (180 - 200 l) made from American white oak, charred where required by regulation. 'White oak' is a cooperage trade classification used for at least 10 oak species, the principal being *Quercus alba*.

Most other types of whisky are matured either in barrels previously used for maturation of Bourbon or in sherry production. Bourbon barrels may be used 'as is' or re-assembled as hogsheads, with additional staves and new ends to increase capacity (250 L). Re-use of casks decreases the levels of wood components extracted during maturation and gives a much less intense wood-derived flavor. Sherry casks used for the

maturation of some whiskies are generally butts or puncheons (480–520 l) which have been used for the fermentation and/or aging of various styles of sherry. Sherry casks can be either American or Spanish oak casks (predominantly *Quercus petraea* and *Quercus robur*). American oak is generally used for fino and amontillado sherries while Spanish oak is used for oloroso sherry. For sherry casks distillers will specify the species of oak, whether the cask is used for fermentation and the contact time with maturing sherry, as all affect the final properties of the mature spirit. In all but Bourbon production, casks are reused until they are exhausted, i.e., fail to produce a satisfactory change in the spirit over an economical maturation period. The life of a cask can be extended by removal of the old char layer and re-firing the cask. This regeneration increases the levels of wood polymer breakdown products extracted during subsequent maturation, but does not regenerate all wood constituents and such casks give different sensory properties when compared with a new cask.

Maturing whisky was traditionally stored in stone built warehouses located beside the distillery. Casks were stored in racks or in ‘stows’, two or three high on the earth floor. Expansion in production resulted in much larger warehouses with steel racking which allowed casks to be stored up to 12 high. The temperature and humidity within warehouses varies between different countries and this can affect the way in which a spirit matures. In cool damp climates, such as Scotland and Ireland, more ethanol than water evaporates from the cask and consequently strength decreases during the maturation period. In warm dry climates, such as the USA, more water than ethanol evaporates and strength increases. During maturation, many of the chemical changes that occur are influenced by the storage conditions, particularly temperature. However though higher maturation temperatures have been shown to increase the rate of extraction of wood components, this does not generally correlate with improvements in sensory quality.

15.3.5 Blending to Bottling

With the exception of occasional “single cask” bottlings, all whiskies are to some extent blended [11,19]. Even in the most tightly controlled environments batch to batch variations can occur at any stage in the production process. Blending is used to produce a consistent product that has the distinctive flavors of that particular brand. For Bourbon and single malt production variations are evened out to create the recognised brand from different batches of whisky. True blended whiskies consist of light-bodied spirits mixed with a number of heavier bodied spirits in a wide range of proportions. ‘Light-bodied spirits’ are those distilled to high ethanol concentrations using continuous column stills and include Scotch grain and American light whiskies. ‘Heavier bodied’ whiskies are either batch pot still products or column still products distilled to lower ethanol concentrations. The nature and components of blends are determined by the traditions and regulations of the country of origin. In Scotland there are a large number of distilleries and blend recipes are complex with as many as 50 malt whiskies used. Purchasing or exchanging new whiskies by producers maintains the complexity of these

blends. In North America, where there are fewer distilleries and trading between competitors is uncommon, the variety of whiskies for blending can be increased by variations in mash composition, fermentation conditions, distillation parameters, maturation periods and cask types.

The physical process of blending is, however, very similar. Approved whiskies are drained from casks into passivated steel troughs, which convey the whisky to a blending vat. Mixing uses mechanical agitators and compressed air. When the blend is correct, de-proofing water is added to reduce the alcohol content to bottle strength. In Scotland blending is sometimes followed by a further maturation or marrying period to allow integration of the individual components into the final product. An additional maturation or finishing period is also used for some single malts. This frequently employs a distinctive wood type, such as port casks, and is used to add complexity to a whisky or to increase the number of styles available from a single distillery.

Most whiskies are filtered prior to bottling to reduce the risk of haze formation during storage. In heavier bodied older whiskies and whiskies matured at high strengths, high molecular weight lipids and wood components slowly flocculate when the whisky is reduced to bottling strength. This is controlled by chill filtration, where whisky is cooled to between -10 and 10°C for a specified period of time and the problem compounds are removed by physical separation and adsorption by a filter. The conventional filter is a plate and frame filter with preformed pads of cellulose, or cellulose impregnated or pre-coated with diatomaceous earth. Operational parameters depend on batch size, the nature of the product and the filtration rate required.

15.3.6 Sustainability

The main risks from climate change for the Scotch whisky industry have been identified as supply and quality of cereals, due mainly to changing precipitation patterns, and a risk to water supplies, again due to changing precipitation [20]. Changes in precipitation have also caused an increase in flood risk [21]. In addition, the processes involved in whisky production are temperature sensitive and may all be affected by temperature change. Similar considerations may apply to other areas of the industry.

The Scottish government has set demanding targets for reductions of carbon emissions [22], and the Scotch Whisky Association has a comprehensive environmental strategy for the Scotch whisky industry [23]. The strategy commits the industry to reducing carbon emissions and reliance on fossil fuels, to manage water requirements, to reduce packaging materials and waste, to seek to ensure that all casks come from sustainable sources, and to work with suppliers to reduce the environmental impact of the industry. The strategy includes a 2050 target of 80% of energy being derived from non-fossil fuel sources.

Environmental performance of a distillery has been defined in terms of a reduction in fossil fuel use, minimisation of water abstraction, and reduction in waste BOD discharged to the environment [24]. A distillery was therefore planned which would be both

fossil fuel and water neutral. This was achieved by integration of the new distillery with an existing maltings. Fossil fuel use was reduced by combustion of solid co-products of the process, and use of methane generated by anaerobic treatment of aqueous co-products. Water abstraction was reduced by treatment and reuse of water.

The environmental impact of cereal supply can be reduced by selection of varieties which give the best alcohol yield with the lowest fertilizer requirement [25,26]. Packaging materials are largely reusable or recyclable, and progress has been made to reduce the weight of packaging used [23]. The other major input into the industry is casks. These are mostly reused casks after Bourbon whiskey maturation, with some Spanish oak casks after their use in sherry production. Timber for Bourbon (and other whiskey) casks is sourced from the white oak forests of the US, and the US Forest Service in Missouri reports that the rate of growth exceeds the rate of harvest [27]. Similarly, the rate of growth in the oak forests of northern Spain exceeds the rate of harvest.

Elsewhere, individual companies have their own environmental and sustainability policies. These are generally similar to the Scotch Whisky Association strategy, though tend to be shorter term and may lack demanding targets. Typically they include most or all of these provisions: to promote sustainable agriculture, and reduce their impact on the wider environment; to reduce non-renewable energy use and greenhouse gas emissions; to reduce water use and improve quality of water discharged to the environment; to reduce packaging material use and ensure it is all reusable or recyclable; to eliminate waste to landfill.

15.4 Composition

With the advancement in chromatographic techniques and, in particular, their coupling with mass spectrometry the number of compounds identified from whisky has increased dramatically [28,29]. Particular use is made of gas chromatography with a number of detectors for the measurement of volatile organic compounds. Non-volatile constituents may be measured as a whole spectrophotometrically, or individually by high performance liquid chromatography. Problems arise, as compounds of sensory importance may be present at very low concentrations (nanograms per gram).

15.4.1 Alcohols, Carboxylic Acids and Esters

Aliphatic alcohols, acids and esters are produced by the yeast (*Saccharomyces cerevisiae*) during fermentation, though ester formation may also occur during distillation and maturation. Saturated and unsaturated chains are found, ranging from one to 16 carbon atoms. Short-chain alcohols such as propanols, butanols and pentanols are the most abundant. Nonaliphatic alcohols include 2-phenylethanol and 2,3-butanediol. The principal organic acid is acetic acid, accounting for between 50% and 90% of the total content of volatile acids. Octanoic, decanoic and dodecanoic acids are quantitatively important, though lower-molecular-weight acids such as butanoic and pentanoic

acids may have more sensory impact. The most abundant esters are the ethyl esters of the acids named, their concentrations reflecting the relative abundance of the acid. Any other combination of the acids and alcohols present may occur. After ethyl esters, the most abundant are acetate esters such as isoamyl acetate and phenylethyl acetate. The esters of hydroxy acids, such as lactic acid, and of dicarboxylic acids, such as succinic acid, can be distilled along with alcohol and steam and so occur in whisky.

15.4.2 Aliphatic Carbonyl Compounds

The majority of carbonyl compounds are produced by the metabolic actions of the yeast during the fermentation. However, their formation by oxidation of unsaturated fatty acids or Strecker degradations has been shown to occur at almost any stage of production. Acetaldehyde is the most common and is often used to determine total aldehydic content. Many other aldehydes, ketones and acetals have also been identified. For aldehydes and ketones, aliphatic chain lengths, both saturated and unsaturated, range from two to 14 carbon atoms. Due to the high alcoholic strengths of the distillate, aldehydes exist in equilibrium with the corresponding acetal and hemiacetal. The most frequently reported acetal is 1,1-diethoxyethane, though a range of aldehydes up to six carbon atoms have been shown to form acetals.

15.4.3 Sulfur and Nitrogen Compounds

Yeasts have the ability to utilise inorganic sulfur, sulphur-containing amino acids and vitamins, and from these sources volatile sulphur-containing compounds are produced. The range of compounds detected includes heterocycles, such as thiazoles and thiophenes, and aliphatic mono-, di- and tri-sulphides. Formation of sulfur compounds has also been shown to occur during distillation and concentrations are highest in new spirit. Cask char, wood extractives and traces of elemental copper are important in the degradation of such compounds during maturation. Heterocyclic nitrogen compounds such as pyrazines, pyrroles and pyridines are formed via the Maillard reaction during cooking, mashing, distillation and barrel charring. As they have relatively low odor thresholds they may well make a contribution to the aroma of whisky.

15.4.4 Phenols

A wide range of phenolic compounds is found in whisky. Simple phenols such as phenol, the isomeric cresols, xlenols, ethyl phenols and guaiacols arise through the thermal degradation of benzoic acid derivatives from malt and from peat smoke. Phenolic aldehydes such as vanillin, syringaldehyde, coniferaldehyde and sinapaldehyde are formed from the breakdown of wood lignin during cask charring and maturation. Their corresponding acids and ethyl esters have also been detected. Polyphenols extracted from the wood include ethanol-soluble lignin and tannins derived from both gallic and ellagic acids.

15.4.5 Heterocyclic Oxygen Compounds

The main types of these compounds found in whisky are furaldehydes and lactones. Furfural (2-furaldehyde) is formed from pentoses and 5-hydroxymethyl-2-furaldehyde from hexoses via the Maillard reaction during mashing, distillation and cask charring. Lactones are formed by the dehydration of aliphatic hydroxy acids and most are derivatives of 2(3H)-furanones. The most abundant of these are the *cis* and *trans* isomers of 5-butyl-4-methyldihydro-2(3H)-furanone, the oak or whisky lactones, which are formed and extracted from oak casks during maturation.

15.5 Variations With Processing

Though there are only four basic processes in whisky production, local, regional and national variations in these processes are responsible for the large number of whiskies available. Each stage of production, from the production of a fermentable extract, through to fermentation, distillation and maturation can affect the character of the final product [30]. However each stage is interlinked and identifying one, as the most important for a particular type of whisky, is often impossible.

15.5.1 Production of a Fermentable Extract

The cereal grains in common use are wheat, corn (maize), rye and barley, though the choice of cereal may be legally restricted. Flavor differences between cereals arise from two quite separate sources: the composition of the extract and the levels of precursors present. The choice of cereal can also indirectly affect the flavor of the final product. The composition of the fermentable extract varies with cereal used and can influence the metabolism of the yeast. Factors such as pH, amino acid concentration and levels of insoluble material are known to affect the levels of fusel alcohols, acids and esters produced which may influence the flavor of the final product. Between cereals differences also exist in the levels of precursors for a number of classes of flavor compounds. Examples are compounds formed by the chemical and enzymatic oxidation of fatty acids; the reaction of free amino acids and reducing sugars (Maillard reaction) and the thermal degradation of biosynthesised precursors like S-methylmethionine and benzoic acid derivatives. However the breakdown of these compounds frequently occurs during fermentation and distillation and is affected by variations in these processes. Consequently few flavors in the final product have been unequivocally linked to components of the original cereal [31].

15.5.2 Fermentation

Fermentation is responsible for the primary production of a large number of flavor compounds in whiskies. Factors that are known to alter the production of these compounds are the strain or mix of yeasts used, design and level of aeration in the fermenter, the duration of the fermentation and the presence of contaminating bacteria.

These factors alter according to distillery design and the raw materials employed and so may be important in producing regional and national differences in whiskies.

15.5.3 Distillation

In batch distillation using pot stills, the selection of the cut points in the still run has a marked influence on the flavor of the product. Generally, aldehyde and short chain ester concentrations are determined by the primary cut from foreshots to spirit, while the concentration of fusel alcohols and acids is determined by the cut from spirit to feints. Other factors, which affect the physical process of separation during distillation, are still design parameters such as the still height, the angle and length of the lyne arm and the type of condenser used. In addition, a number of chemical reactions can occur when the wash or low wines are heated for distillation. This is particularly true for batch distillation in pot stills where the heat impact is greater. Examples include the formation of sulfur compounds from sulphur-containing amino acids, the breakdown of unsaturated fatty acids to carbonyl compounds, and the dehydration of β -hydroxypropionaldehyde to acrolein. Copper catalyses many of these reactions.

In continuous stills congener concentrations are determined by the design, operation and efficiency of the still. The range of reactions described for pot stills can still occur but become progressively less important as distillate strength increases. Collection of spirit at high ethanol concentrations (>90%) results in low levels of congeners in the spirit.

15.5.4 Maturation

During the maturation period the new distillate becomes highly modified as a result of its contact with the cask [31]. The composition of the cask wood has a major effect on the extent of this modification and varies with the species of oak used, the pre-treatment applied to the wood and the previous use of the cask. In general, American oak (*Quercus alba*) yields higher concentrations of vanillin and *cis* oak lactone, while Spanish oak (*Quercus robur*) gives higher concentrations of tannins. Charring increases the levels of lignin breakdown products and lactones extracted during maturation, whereas previous use of a cask greatly reduces the amount of extractable lignin breakdown products and lactones. The use of sherry casks results in raised levels of tannins and sugars. The cask is the major determinant of mature quality, but other factors such as filling proof, climate and warehouse conditions also affect the course of maturation. Higher temperatures produce greater evaporation rates and also increase extract levels and some reaction rates, but do not necessarily produce a more mature product.

15.6 Sensory Assessment

Whisky has traditionally been assessed by expert blenders with many years of experience and training within the industry. An experienced blender knows what flavors of distillate a still can produce, which are desirable, and how a whisky is likely to develop during

maturation. The blender's task is then to identify faults and deviations from the expected path of maturation, and to select a specific maturation point at which a whisky can contribute to the blend. Blenders use a system of flavor description to assist their work, but their method of working is largely by comparison of samples with experience or a reference. Each sample is evaluated in terms of its similarity to an expected or acceptable product [30,32].

Scotch grain whiskies are light-bodied continuous still spirits, largely regarded as providing a background for a blend. Malt whiskies provide the majority of the character of a particular blend and, though the distilleries all produce slightly different whiskies, for convenience they are divided into groups sharing common characteristics. The Highland malts, the most numerous, are further divided into first, second and third class. These are not intended to be quality evaluations, but are simply descriptive classes. The Lowland malts are not regarded as having a distinctive character, and thus can be used in relatively large quantities in blends. There are three remaining Campbeltown distilleries and this group has, as a result, declined in importance. The Islay whiskies tend to be full flavored and are usually heavily peated, and thus have a significant impact on a blend even when used in small proportions. The Highland group contains most of the great whiskies which have established reputations as single malts and for their value in blending. A wide range of flavors is available and a skilled blender can make use of them to create a distinctive blend.

Formal methods of laboratory sensory analysis have also been used. Many companies use difference tests for quality control monitoring, and the whisky industry seems to have been one of the first to adopt the duo-trio and triangle tests for such purposes. Over recent years there has also been an increasing interest in the assessment of whiskies by profiling or descriptive methods, largely for research into the origins and development of flavor in whiskies. Several broadly similar vocabularies of descriptive terms have been used. These vary mainly in the number of terms they include, and therefore in the detail with which they attempt to describe flavor. They have often been presented in the form of a "flavor wheel", with varying numbers of concentric tiers, which while having no particular theoretical significance provide a framework for communication and training [33].

15.7 Correlation of Chemical Composition With Sensory Data

The flavor of whisky is dependent upon a complex mixture of components that originate from and are modified by different stages of production [30]. While improved analytical methods have revealed much about the chemistry of whisky, this has not resulted in a better understanding of the chemical basis of whisky flavor and quality. Frequently chemical differences can be related to variation in production processes, but identifying the actual compounds responsible for the flavor differences has been much more elusive.

The extent to which volatile compounds from cereals survive fermentation, distillation and maturation is the hardest to establish. Whisky prepared from heavily peated malt retains a characteristic medicinal, smoky flavor that is strongly correlated with the total phenols content of the malt and of the whisky. However the range and concentration of phenols present is greatly altered by mashing, fermentation and distillation and whether the phenols fully explain the 'peated' flavor is still uncertain. Likewise, some whiskies (especially Canadian) have a characteristic 'spicy' or 'minty' flavor, which is possibly related to the use of rye in the mash, but no compound has been identified as responsible.

Various flavors in new distillates have been correlated with sulfur compounds. Polysulphides are present in whisky at levels around and above their sensory thresholds (3–6 ng/g for dimethyl trisulphide), and the data available suggest that these compounds are correlated with 'sulphury' and 'heavy' characters in fresh distillates. Other sulphur-containing compounds are thought to be important contributors to flavor and their formation is dependant on the levels of precursors present in different cereals/cultivars. However such compounds are frequently formed during fermentation and distillation and then degraded during maturation. Each stage has its own impact on the concentration in the final product making it harder to establish their role in whisky flavor.

The products of fermentation and distillation make a substantial contribution to whisky flavor. However these compounds, except when present in excess, appear to form what has been described as the 'integrated whisky complex', and their individual contributions have not been determined. It does appear that relatively constant proportions of some of these compounds must be present for a recognisable 'whisky' character, though substantial changes can occur with only minor changes in flavor.

Compounds considered to be important in maturation are vanillin, eugenol and *cis* oak lactone. These have been correlated to 'vanilla' and 'clove-like' flavors in some whiskies matured in new casks. However cask reuse causes a general decrease in wood extractives and their importance in whisky matured in refill casks is still uncertain.

Attempts to create a model of whisky flavor have been largely unsuccessful. Attempts were based on compositional analyses of the liquid. Research has since shown that there are complex relationships governing the release of flavor compounds from whiskies. Major factors identified include alcoholic strength and the non-volatile content of the whisky matrix. Dilution of spirits for sensory assessment, to reduce the pungency of ethanol, can greatly alter the release of aroma compounds more soluble in ethanol than in water. Dilution also increases the capacity for matrix effects that suppress the release of ethyl esters and other aroma compounds. Matrix effects may occur through competition for release at the liquid surface or changes in solubility of flavor components. Changes in solubility can be related to the physical structuring of the ethanol-water mixture. Differential scanning calorimetry, light scattering and mass spectrometry have all demonstrated that physical changes occur during maturation, adding another layer of complexity to whisky flavor and reinforcing the uniqueness of each individual product.

15.8 Conclusions and Perspectives

The whisky industries world-wide are generally mature and technically stable, with a well understood and well controlled process. The composition and process for traditional products have been stabilised by legislation, but leaving scope within the regulations for limited innovation. Mature markets are in some cases showing a slow decline in consumption, offset by increasing consumption in less traditional markets. There are particular opportunities for premium spirits such as whisky in emerging economies, where local products may be partially displaced. Major challenges for the industries will be the development of more sustainable processes and procedures, in the face of increased demands for reductions in fossil fuel use, water abstraction and waste production. A further challenge is in combating counterfeit spirits. Any premium product or brand can command a price advantage over other products, and this may attract some producers to attempt to pass off inferior or counterfeit products as a genuine product or brand. Analytical procedures must be used to demonstrate that a suspect product is not what is claimed, and cooperation with regulatory authorities in the country of origin is a necessary route to halting their production and distribution.

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